

MARCO FERNSTAEDT

Phoenix, AZ | 928 256 9277 | Fernstaedtmarco@gmail.com

GitHub: <https://github.com/MarcoFernstaedt> | LinkedIn: <https://www.linkedin.com/in/marco-f-19a372219> | Portfolio: <https://marcofernstaedt.com>

SOFTWARE DEVELOPER

SUMMARY

Software developer focused on React, Next.js, TypeScript, Node.js, Express, MongoDB, API integrations, and production web delivery.

Builds full stack applications with authentication, WebSockets, file uploads, AI API integrations, SEO, analytics, responsive interfaces, and clear documentation. Customer service background supports user focused debugging, direct communication, and practical product thinking.

CORE SKILLS

JavaScript, TypeScript, React, Next.js, Vite, Node.js, Express, MongoDB, Mongoose, REST APIs, WebSockets, Socket.IO, JWT authentication, Clerk, Stream Video, Stream Chat, Monaco Editor, Piston API, OpenAI APIs, Resend, Cloudinary, Tailwind CSS, HTML, CSS, Git, GitHub, Vercel, Linux, DNS, SSL, accessibility, debugging, documentation

SOFTWARE DEVELOPMENT EXPERIENCE

Independent Software Developer | Phoenix, AZ | 2025 to Present

- Built and deployed production style software projects across Next.js, React, TypeScript, Node.js, Express, MongoDB, OpenAI APIs, Resend, Vercel, and Linux based development environments
- Delivered a production tutoring business website for Saguaro Blossoms Learning with responsive UI, service landing pages, SEO metadata, Schema.org JSON LD, contact intake, Resend email delivery, analytics events, and custom domain deployment
- Built and maintained a public developer portfolio using Next.js, TypeScript, Tailwind CSS, Framer Motion, Vercel Analytics, GitHub metadata APIs, structured project data, a resume download asset, and responsive project modals
- Created developer facing documentation, proof notes, deployment checks, and repeatable verification steps for portfolio, client site, and application work
- Used Git, GitHub, browser debugging, API testing, build verification, and production deployment checks to ship and validate changes across multiple projects

PROJECTS

AI Image to Audio Accessibility Tool

GitHub: https://github.com/MarcoFernstaedt/image_accessibility_tool

Live: <https://ita-orpin.vercel.app>

- Built a Next.js and TypeScript image accessibility platform that converts uploaded images into concise spoken descriptions for blind and low vision users
- Implemented an OpenAI Vision to text to speech pipeline using image analysis, generated descriptions, MP3 audio responses, and browser Blob playback and download flows
- Engineered secure image ingestion with multipart FormData parsing, MIME validation, 10 MB file limits, base64 Data URL conversion, and Node.js Buffer based audio streaming
- Added accessibility and production controls including semantic labels, aria live status messaging, keyboard friendly UI, Arcjet shield rules, bot detection, hosting IP checks, and token bucket rate limiting

Real Time Messaging Platform

GitHub: https://github.com/MarcoFernstaedt/socketio_chat_app

- Developed a full stack real time chat application with React, TypeScript, Node.js, Express, MongoDB, Mongoose, and Socket.IO for authenticated one to one messaging
- Implemented secure authentication with bcrypt password hashing, JWT HTTP only cookies, protected REST endpoints, safe user serialization, and authenticated WebSocket handshakes
- Built real time messaging infrastructure with online presence broadcasts, direct message socket delivery, connection handling, disconnection handling, and MongoDB backed conversation history
- Created a responsive chat frontend using Zustand state management, optimistic UI updates, image message support through Cloudinary, notification sounds, contacts filtering, and chats filtering

InterviewLab Code Live Platform

GitHub: https://github.com/MarcoFernstaedt/code_live_platform

- Built a live coding interview platform with React, TypeScript, Express, MongoDB, Clerk authentication, Stream Video, Stream Chat, and Monaco Editor
- Designed protected session APIs for creating, joining, listing, and ending coding sessions with host and participant authorization, active and completed lifecycle state, and MongoDB persistence
- Integrated Stream Video and Stream Chat so each coding session provisions a dedicated video call and chat channel, adds participants on join, and tears down resources when the host ends the session
- Implemented an in browser coding workspace with Monaco Editor, resizable panels, JavaScript, Python, and Java support, and Piston API code execution with compile error, runtime error, and timeout handling

Saguaro Blossoms Learning Client Website

GitHub: <https://github.com/MarcoFernstaedt/cynthia-tutoring-platform>

Live: <https://saguaroblossomslearningservices.com>

- Built a production Next.js, TypeScript, and Tailwind CSS client website for a tutoring business with responsive pages, mobile navigation, bilingual English and Spanish content, pricing sections, and service conversion flows
- Implemented a server side contact API using Next.js Node runtime and Resend, including required field validation, email validation, honeypot spam prevention, HTML escaping, owner notification emails, and visitor confirmation emails
- Delivered SEO infrastructure with canonical metadata, Open Graph and Twitter previews, sitemap and robots generation, service specific landing pages, and Schema.org data for organization, local business, services, person, and FAQ entities
- Added Vercel Analytics event tracking for language switches, consultation clicks, service clicks, pricing views, navigation, SEO topic clicks, contact form start, and contact form submit events

EDUCATION AND TRAINING

Self directed software development training through production style projects in React, TypeScript, Next.js, Node.js, MongoDB, authentication, APIs, real time systems, AI integrations, accessibility, and deployment workflows

CompTIA Network Plus in progress